

A Correlated Signals

Computational Game Theory

Model of Litigation Bargaining

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Abstract

A linear programming algorithm solves extensive-form game trees representing litigation games in which parties receive correlated quality signals. Model permutations allow for asymmetric two-sided information, risk aversion, costs, and information quality, while providing litigants with file, answer, offer, and quit decisions. This allows examination of interactions among different game features and parameters. With baseline parameters, including risk neutrality, the parties are strikingly aggressive in their settlement demands, and settlement is relatively rare. The effects of fee shifting depend markedly on whether fee shifting is triggered only at trial or also by nonanswer or pretrial withdrawal.

1 Introduction

A natural assumption for settlement bargaining models is that the plaintiff and defendant receive correlated signals of the quality of the litigation. In a tort case, for example, each party may possess a combination of shared and private information about the events leading to an accident, and so each party can be understood as receiving a noisy signal of the true merits of the case. Even in a case in which facts are unambiguous and the only dispute is as to the law, each party will assess the case based on its lawyer's analysis and hunches, and the parties' estimates of case strength based on such legal advice will be correlated but not necessarily identical. Most of the literature modeling settlement bargaining, however, does not feature correlated signals. In the classic models of Landes (1971) [13], Posner (1973) [17], and Gould (1973) [9], parties' estimates of

litigation quality are entirely exogenous. Later Bayesian models, such as Bebchuk (1984) [3] and Polinsky and Rubinfeld (1998) [16], assume that one party knows the quality of litigation while the other party knows the distribution of all cases but does not receive a signal of the particular case's quality.

Instead of correlated information, some models grant the plaintiff some information and the defendant some other information. The aggregate of these pieces of information then determines case quality. For example, Daughety and Reinganum (1994) [7] assign one party information on liability and another party information on damages. Similarly, in Friedman and Wittman (2006) [8], Dari-Mattiacci and Saraceno (2020) [6], and Abramowicz (2025) [1], the plaintiff and defendant each receive independent random signals from a known distribution, and the true value of the case depends on an average of those signals. Thus, in these models, each of the plaintiff's and defendant's signals correlates with the merits, but unrealistically, the signals do not correlate with one another. The game thus becomes like a game of poker in which there are no common cards.

Plausibly, the optimal strategy and resulting equilibria in such a game might be quite different from those in a game of correlated signals. An intuition is that correlated signals would lead to high rates of settlement and relatively limited opportunities for strategic negotiation behavior. With independent signals, each litigating party must worry greatly about adverse selection; if the other party agrees to settle, that may well be because the other party has received a signal that the first party's claim is actually quite strong. But with correlated signals, there is only a small probability that the other party has a highly divergent view of the case strength. A reasonable strategy then might be to make demands or offers that rise roughly linearly with case strength, effectively agreeing roughly to share the surplus from settlement. Settlement might still fail because of tail estimates, but cases that do fail should resolve at roughly their expected value.

Only one pair of articles allows the plaintiff and defendant to estimate the quality of the litigation based on correlated signals. Lee and Klerman (2016) [14] and Klerman, Lee, and Liu (2018) [12] use correlated signals to update the canonical selection model of Priest and Klein (1984) [18]. In these updated models, the true value of the litigation is Y (distributed uniformly over the entire real line), and the plaintiff and defendant receive signals of $Y + \epsilon_p$ and $Y + \epsilon_d$, respectively, where each ϵ is a normally distributed random variable. The papers differ in that the older adopts the Landes-Posner-Gould assumption that the cases settle whenever there exists a settlement amount at which each party prefers to settle, while the more recent explicitly models the bargaining structure. The later paper's bargaining model is that of Chatterjee and Samuelson (1983) [4], in which each party separately makes an offer and settlement occurs at the midpoint if the plaintiff's offer is less than or equal to the defendant's.

These papers represent significant advances in the realism of litigation models. But they have limitations that make it difficult to assess how correlated signals would impact issues other than the selection of cases

for litigation. First, even in the more recent paper, the litigation quality is constrained to a range where each party has a credible threat to take the case to trial (pp. 392-93). In effect, the plaintiff always files, the defendant always answers, and neither party ever quits before trial. Yet we have known since Shavell (1982) [19] and P'ng (1983) [15] about the importance of filing and answer decisions and since Huang (2004) [10] about the strategic importance of the plaintiff's option to abandon a suit. Second, the entire cost of litigation is thus borne all at once, rather than sequentially. Yet expenditures on litigation are dispensed over time. Third, although case quality varies, there is a simple cutoff, above which the plaintiff has a high fixed probability of winning and below which the probability is fixed at a low value. More realistically, the probability of winning may be close to 50% in some cases and a range of high or low values in others.

In this article, I relax these assumptions through use of a computational model that identifies Nash equilibria in large extensive form game trees, applying the linear programming algorithm identified in von Stengel, van den Elzen, and Talman (2002) [20]. Abramowicz (2025) applies this algorithm to model settlement bargaining but does not consider correlated signals of litigation quality. Rather, that article seeks to establish a proof of concept for computational (or algorithmic) game theory modeling by using a setup similar to that of Dari-Mattiacci and Saraceno, in which the parties have independent signals that aggregate to determine case quality. The ten-offer specifications in the present article have 45,211 nodes, while the fifteen-offer extension has 97,211 nodes.

Despite this greater complexity, the solutions identified in this article remain perfect Bayesian Nash equilibria. That is, the plaintiff and defendant each assign a mixed strategy to each information set, and each player's mixed strategy is a best response for each player given that player's information. That is, neither player could improve its strategy even by a small ϵ by changing its strategy, holding the other player's strategy constant, in the modeled game. Perfection guarantees that a player's choices remain optimal if a decision is actually reached, even at decision points that are not reached in equilibrium, given beliefs satisfying the equilibrium's consistency requirements. The mixed strategies are exact rational numbers, although the payoffs are rounded before the game is solved.

The algorithmic game theory model thus follows in the Bayesian tradition of recent settlement bargaining scholarship, but with enhanced strategic complexity, including multiple levels of case quality, multiple signal levels, variable levels of fee-shifting, adjustable levels of risk aversion, and litigant options to file, answer, and quit before trial. The richness of game features can provide a useful complement to analytical models. The computational approach shines in three ways. First, it can illustrate how different game features may interact. Second, the computational approach can make it possible to explore relatively small changes in applicable rules. Third, welfare effects may be precisely measured, and different aspects of accuracy may be distinguished.

The comparison of fee-shifting rules illustrates all of these advantages. Switching from our baseline American rule simulation to a regime in which fees are shifted after trial makes the plaintiff more conciliatory at one signal consistent with a slightly weak case, producing an increase in settlement but leaving the rest of the equilibrium largely intact. Extending fee shifting to cases in which the defendant does not answer or a party withdraws, however, has a much more substantial effect. The direct effect of the change is that a defendant with a weak case is paradoxically more likely to answer, because a defendant who does not answer must not only pay damages, but also the plaintiff's filing fees. That in turn discourages plaintiff filing. Expansion of fee shifting thus adversely affects meritorious plaintiffs while benefiting truly not liable defendants.

This is not a just so story, but a set of conclusions that can be supported by considering equilibria and how each party's best response would change if only parts of the opponent's strategy changed. Admittedly, a computational solution can lack the elegance of a mathematical proof. Indeed, the great virtue of being able to test how different decisions and rules interact will tend to mean that conclusions will be guarded. Still the model not only reveals an intuition, that extending fee shifting to nonanswer makes defendants more willing to answer, but also can show that the magnitude of this effect may outweigh competing effects. This analysis does not conclusively resolve the question of what stages should be included under a fee-shifting rule, let alone the broader debate concerning the advisability of the American and British rules, but it helps identify some of the policy stakes.

These results relate to a literature showing how the timing of litigation expenditures shapes bargaining by changing the credibility of threats. Bebchuk (1996) [2] shows how dividing costs across stages of litigation can make what would be a negative-expected-value claim if fully litigated viable, while Croson and Mnookin (1996) [5] and Hubbard (2016) [11] show how litigants can enhance the credibility of the threat to continue pursuing a claim or defense by sinking expenditures in advance. Central to this logic is the option to exit before trial. The greater the extent to which a party has already incurred costs before the exit decision and thus reduced the costs remaining to be incurred, the less attractive exit becomes. That in turn makes the party's continued litigation more credible in earlier settlement bargaining.

The wrinkle added by extending fee shifting to exit is that if exiting itself would force a party to pay opponent's costs incurred to date, that also ultimately makes the party's continued litigation more credible. This can counteract the strategic advantage that the plaintiff may otherwise gain from incurring costs before the defendant. When the plaintiff spends money to file but a fee shifting rule means that the defendant might bear that cost if it does not answer or exits before trial, then by answering, the defendant reduces the chance that it will become liable for those expenses, while meanwhile, its own expenses in answering might end up being borne by the plaintiff. The option of exiting before trial, meanwhile, become less attractive

to both parties, since exiting concedes the other party's filing or answering, as well as in pretrial litigation. While it is not entirely intuitive how much these effects will matter, in the model, they end up being of vital significance. While plaintiffs often do not file, defendants always answer, in both the risk neutral setting and with risk aversion, including in robustness checks assessing the possibility of multiple equilibria. It is this dynamic that ultimately redounds on average to the defendants' benefit with complete fee shifting.

The computational results also offer many intriguing observations along the way. The parties are highly aggressive bargainers with risk neutrality, not so much deciding how much of the perceived settlement surplus they wish to insist on, but instead holding out for outcomes far more favorable than their assessments of case strength would suggest. Offers and demands often remain constant across multiple signal levels before jumping at a threshold. Substantial bargaining failure occurs even though parties have relatively good information and even though the one-round structure eliminates opportunities to influence an opponent's offer through one's own. Meanwhile, equilibria with greater settlement do not necessarily offer lower expenditures or greater accuracy. Finally, risk aversion can increase settlement while also increasing expenditures and causing settlement values to depart further from the amounts warranted by underlying liability, particularly in the absence of fee shifting.

2 The Model

2.1 Information structure

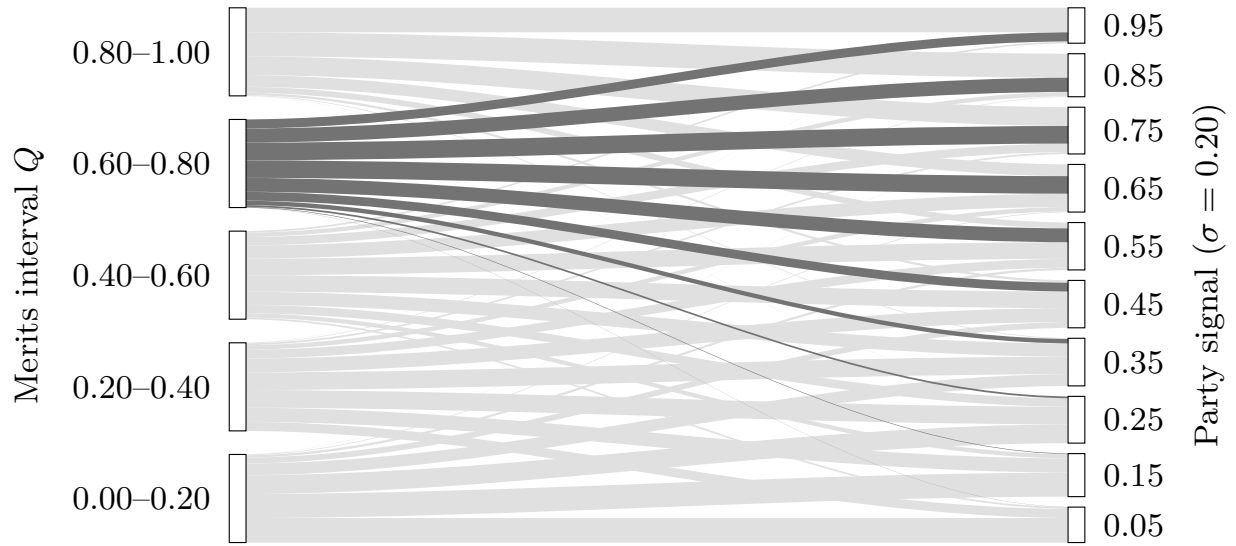


Figure 1: Information structure

2.2 Litigation sequence and updating

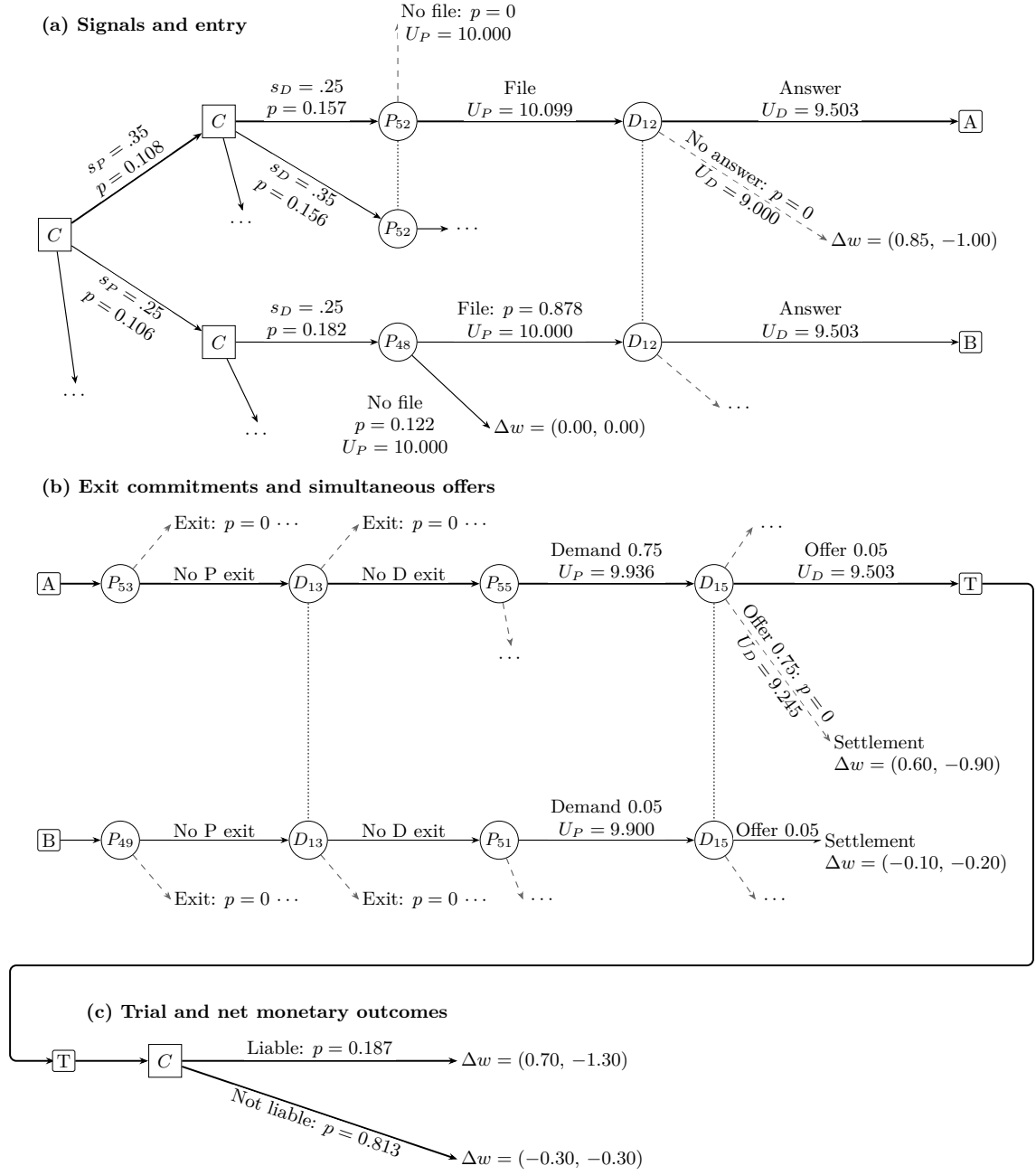


Figure 2: Worked equilibrium path

2.3 Costs, preferences, and fee rules

Table 1: Model primitives

Primitive	Core comparison			
Common merits	$Q \sim U[0, 1]$; $T \mid Q \sim \text{Bernoulli}(Q)$; $\Pr(T = 1) = 0.5$			
Private information	One noisy merits signal per party; 10 signal bins each			
Signal noise	$\sigma_P = \sigma_D = \sigma_C = 0.20$			
Adjudication	Two court-signal bins; fixed damages $d = 1$			
Initial wealth	$w_P = w_D = 10$			
Filing and answering	Endogenous; cost 0.15 for each acting party			
Trial	Additional cost 0.15 per party; no additional bargaining cost			
Exit	Parties commit before offers to continue or exit if bargaining fails			
Bargaining	One simultaneous round; 10 offers 0.05, 0.15, $\dots$, 0.95; midpoint settlement if demand $\leq$ offer			
Preferences	Risk neutral or symmetric CARA, $\alpha = 2$			
Cost comparisons	Common cost multipliers 0.25, 0.5, 1, 2, 4			
Integration	64-point Gauss–Legendre integration over continuous merits			
		American	Trial Fee-Shifting	Complete Fee-Shifting
Fee trigger				
After trial		Own costs	Loser pays	Loser pays
Initial nonanswer		Own costs	Own costs	D pays P’s fees
Later P abandonment		Own costs	Own costs	P pays D’s fees
Later D default		Own costs	Own costs	D pays P’s fees
Settlement		Own costs	Own costs	Own costs

2.4 Equilibrium and computation

2.5 Outcome measures

3 Correlated Signals and Fee Shifting: Risk-Neutral Results

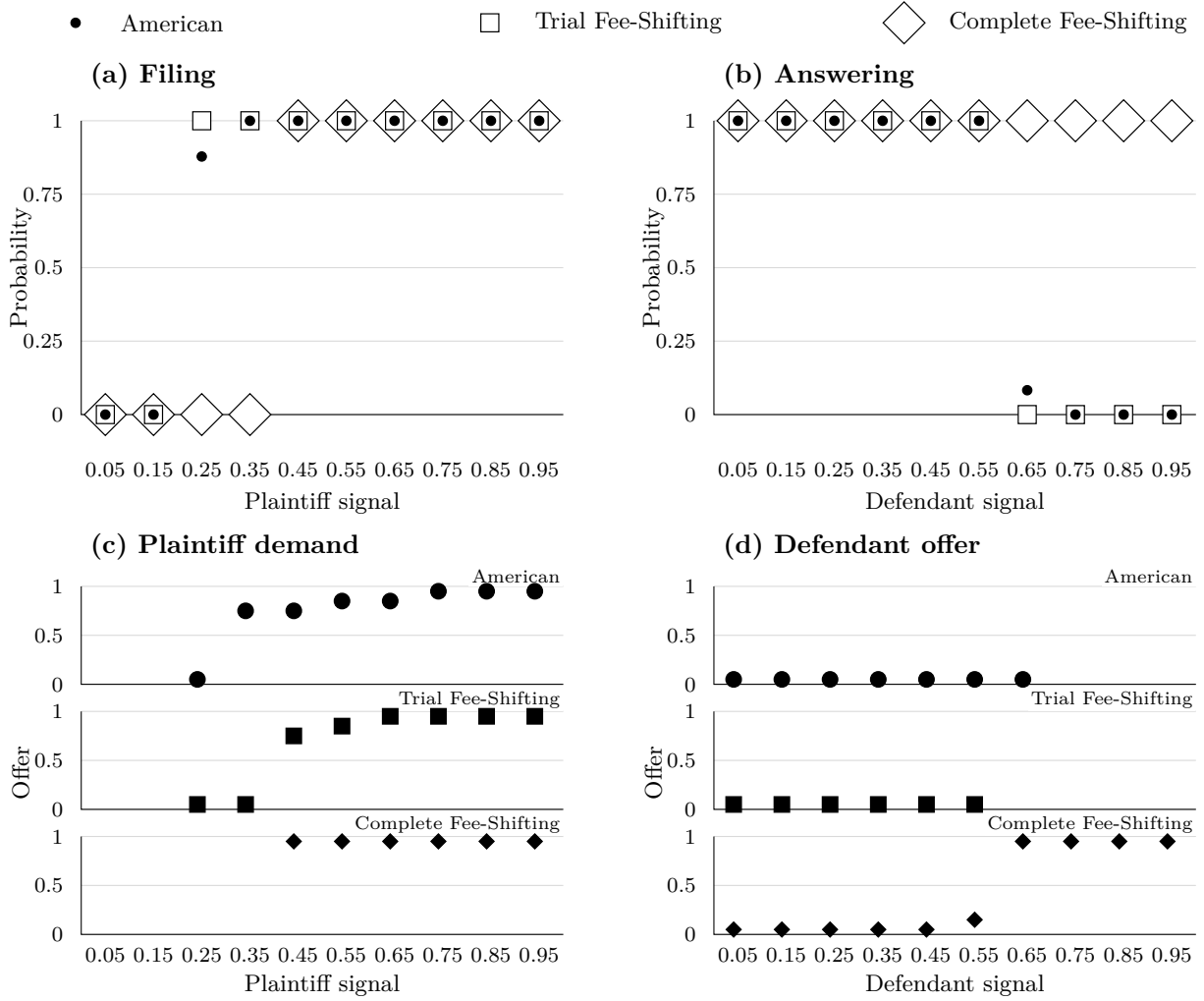


Figure 3: Participation and offers

3.1 Aggressive bargaining under the American rule

3.2 Trial Fee-Shifting

3.3 Complete Fee-Shifting and the incentive to answer

3.4 Defense, filing, and the bargaining population

3.5 Monetary consequences

Table 2: Welfare outcomes

Risk neutral					
Fee rule	Meritorious-P shortfall	Nonliable-D burden	Liable-D excess burden	Gross outcome error	Total expenditures
American	0.216	0.222	0.028	0.278	0.292
Trial Fee-Shifting	0.209	0.198	0.047	0.277	0.266
Complete Fee-Shifting	0.232	0.165	0.075	0.285	0.259
Risk averse					
Fee rule	Meritorious-P shortfall	Nonliable-D burden	Liable-D excess burden	Gross outcome error	Total expenditures
American	0.256	0.256	0.011	0.344	0.336
Trial Fee-Shifting	0.224	0.164	0.017	0.286	0.183
Complete Fee-Shifting	0.241	0.166	0.038	0.298	0.225

4 Risk Aversion and Litigation Outcomes

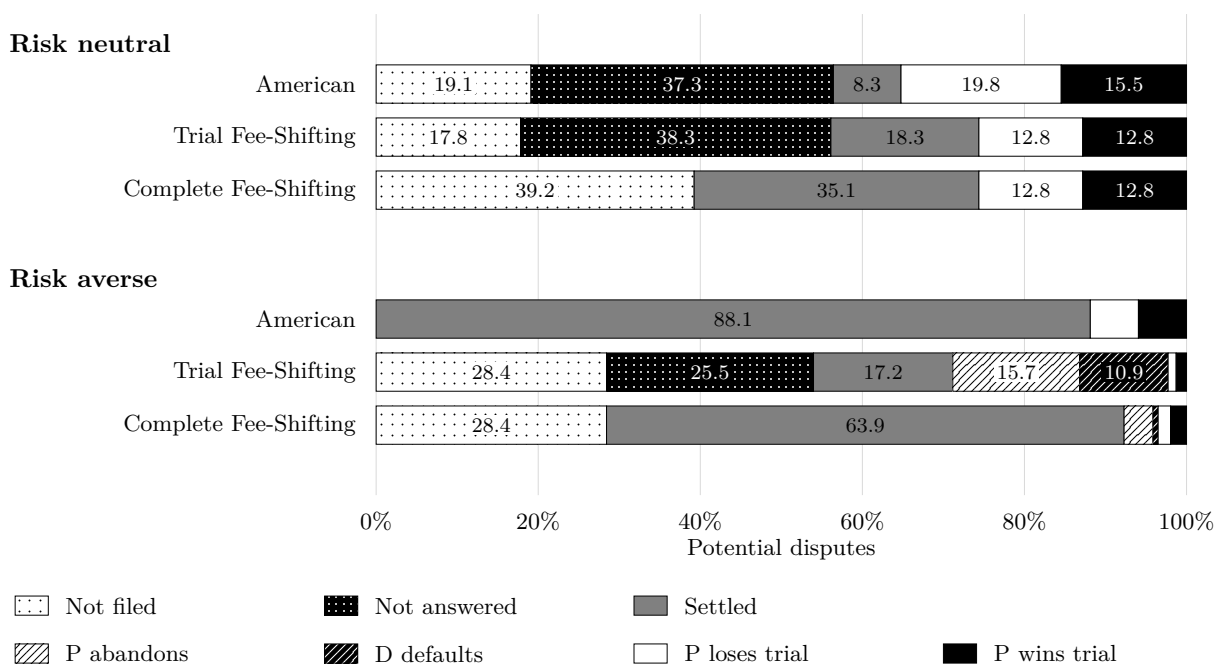


Figure 4: Dispositions

4.1 Settlement, participation, and expenditures

4.2 Risk aversion under the two fee-shifting rules

Table 3 (Strategy mechanisms): to be reduced before insertion.

4.3 Monetary burdens and the cost sweep

4.4 Legal implications

5 Extensions, Robustness, and Limits

5.1 Information quality, merits, and cost timing

5.2 Multiple equilibria

5.3 Finer offer grid

5.4 Empirical interpretation and model limitations

6 Conclusion

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- A Model and computational details
- B Alternative information structures
- C Costs, extensions, and strategy mechanisms
- D Equilibrium and discretization checks
- E Data and reproducibility